

LECTURE-21

Wednesday
22/4/26.

Confidence Interval for Pop. Mean (μ)

σ - Known

Z - Test

$$= \bar{X} \pm Z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}}$$

σ - Unknown.

$n \geq 30$

Z-test

$$= \bar{X} \pm Z_{\alpha} \cdot \frac{S}{\sqrt{n}}$$

SD of Sample

$n < 30$

T-test

$$= \bar{X} \pm t_{\left(\frac{\alpha}{2}, (v)\right)} \cdot \frac{S}{\sqrt{n}}$$

$v = n - 1$

$\alpha = 1 - CL \rightarrow \alpha/2 = \text{answer}$

Now look for answer in t-table against $(v = n - 1)$

Biased SD for Sample (Ztest)

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$$s = \sqrt{\frac{1}{n} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right]}$$

Unbiased SD for Sample (T-Test)

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

$$S = \sqrt{\frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right]}$$

Q: Duration of watch time of Youtube videos is monitored. A random sample of 20 have average time of 15 Hours with $SD = 5$ hours.

(a) Construct 90% C.I for watchtime average.

$$n = 20, \bar{x} = 15, s = 5$$

* Since σ is Unknown and $n < 30 \therefore$ T-test with SD of sample = 5.

$$\alpha = 1 - 0.90 = 0.10$$

$$\alpha = 0.1$$

$$\alpha/2 = 0.05$$

$$v = n - 1$$

$$v = 20 - 1$$

$$v = 19$$

* value of T-test from T-table with these α & $v = 1.729$

$$15 - (1.729) \cdot \frac{5}{\sqrt{20}} < \mu < 15 + (1.729) \cdot \frac{5}{\sqrt{20}}$$

$$13.07 < \mu < 16.93 \rightarrow \text{answer.}$$

(b) If Past Study shows $SD = 6.5$ hours. Find 95% C.I

* Since SD of "past study" $\therefore$ it's $\sigma = 6.5$ (of population) and we gonna use Z-Test

$$\alpha = 1 - 0.95$$

$$= 0.05$$

$$\alpha/2 = 0.025$$

$$1 - \alpha/2 = 0.975$$

$$\rightarrow Z = 1.96$$

$$\mu = 15 \pm (1.96) \cdot \frac{6.5}{\sqrt{20}}$$

$$12.15 < \mu < 17.85$$

$\Rightarrow$ Formulation of Hypothesis:

• Null Type $\rightarrow H_0: \mu = \text{original statement in question (like } \mu = 15 = H_0 \text{ above)}$

• Alternate $\rightarrow H_1: \mu \neq, <, >$ the original one in question (like: $\mu < 15, \mu > 15, \mu \neq 15$)

* H_1 has no need in calculations; But it just tells if H_0 rejected then defo. H_1 is right.

$\Rightarrow$ Types of Errors:

• Type I Error: Rejecting H_0 when it is TRUE.

• Type II Error: Accepting H_0 when it is FALSE